

Hurricane Relief & Response Application

Software Requirements Specification



CSCE 247: Software Engineering

Aisha Patel, Ashley Llontop, Leighton Keenan, Clayton Porter, Savannah Allen

August 29th, 2026

Version 1.0

Table of Contents

Hurricane Relief and Response Application	1
Aisha Patel, Ashley Llontop, Leighton Keenan, Clayton Porter, Savannah Allen	1
1. Introduction	3
2. Stakeholders	4
3. Constraints	9
4. Overall Description	9
5. Business Use Cases	10
6. Functional Requirements	11
7. Non-Functional Requirements	11
9. Competitive Analysis	12

1. Introduction

Purpose

Coastal communities along the Atlantic are increasingly threatened by severe weather, leaving millions of residents at risk of being displaced and facing long-term shortages of resources during major storms. In times of crisis, it can be chaotic and slow to find reliable information about shelter availability, open supply distribution points, and active rescue routes. This is why we aim to develop a mobile app that will help reduce panic and logistical issues that come with emergency responses.

Our app will have a user-friendly interface that lets users quickly access real-time relief options and identify the safest resources nearby. It will gather and compare information on shelter capacities, medical station availability, clean water distribution points, volunteer opportunities, and real-time road hazard alerts to help affected residents, emergency responders, local business owners and volunteers find the best immediate options.

Scope

This document will cover:

- The personas of potential users, including residents, emergency responders, and volunteer organizations.
- Any technical, environmental, or limitations affecting this project.
- An analysis comparing current disaster management systems to highlight the benefits and features of this project.

2. Stakeholders

1. First responders – First responders need quick and accurate information about outcomes from hurricanes, people that need help, road conditions, and emergencies to accurately help everyone efficiently.
2. Insurance Companies – Insurance companies need information about property damage from the hurricane and accurate information about claims so they can determine their losses and if they are legitimate.
3. Volunteers – Volunteers need to know where to help, what supplies are needed, and who needs shelter and food provided so that they can assist people that were affected by the hurricane.
4. Hurricane Victims – Hurricane Victims need access to information about where to evacuate and the closest shelter locations, emergency alerts, and a way to request for help.

Personas

Jasmine Carter

Age: 40

Hometown: Houston, TX

Occupation: Disaster Relief Coordinator

Family: Married, 1 child

Organization: American Red Cross

Role: Relief Organization Worker



Personality Trait

Introvert --- Extrovert	More Extrovert
Thinking --- Feeling	More Feeling
Sensing --- Intuition	Slightly Sensing
Judging --- Perceiving	Strongly Judging

Bio: Jasmine is a disaster relief coordinator and mother who enjoys volunteering, traveling, and spending time with family. She helps organize shelters, volunteers, donations, food, water, and other emergency resources. She is organized, compassionate, and enjoys working with others to solve problems.

Goals: Jasmine wants to see which communities need the most resources so she can distribute supplies and volunteers effectively. She wants to keep track of available resources, volunteer assignments, and requests for assistance. She also wants to communicate with residents and other organizations to make sure everyone is working with the same information.

Frustrations: Sarah dislikes having to gather information from multiple organizations and systems. She gets frustrated when she can't easily determine where resources are needed the most. She also finds it difficult to keep track of large numbers of volunteers, requests, and supplies when information is constantly changing.

James Rodriguez

Age: 38

Hometown: Miami, Florida

Occupation: Firefighter

Family: Married, 2 children

Role: Law Enforcement



Personality Trait

Introvert --- Extrovert	More Extrovert
Thinking --- Feeling	Strongly Thinking
Sensing --- Intuition	Strongly Sensing
Judging --- Perceiving	Strongly Judging

Bio: James is a firefighter and father of two who enjoys fishing, cooking, coaching his son's basketball team, and spending time with his family. He is calm under pressure, observant, and loves to serve his community.

Goals: James wants to quickly see reports from residents and identify which situations require immediate attention. He wants access to updated information about road closures, dangerous locations, and emergency reports. He also wants to communicate with other responders so they can coordinate their efforts.

Frustrations: James dislikes receiving incomplete reports because they make it harder to determine which situations are serious. He gets frustrated when information from different agencies is difficult to access. He also needs information to update quickly because emergency situations can change without warning.

Sarah Sterling

Age: 32

Hometown: Charleston, SC

Family: Married, 1 Child

Occupation: Middle School Teacher

Role: Hurricane-Affected Resident



Personality Trait

Introvert --- Extrovert	Slightly Extrovert
Thinking --- Feeling	More Feeling
Sensing --- Intuition	More Sensing
Judging --- Perceiving	More Judging

Bio: Sarah is a middle school teacher and mother who enjoys spending time with her family, cooking, hiking, and being involved in her community. She is caring, organized, and values having reliable information when making decisions.

Goals: Sarah wants to be able to quickly find shelters, food, water, and medical assistance for her family. She wants to know which areas are safe and receive updates about road closures and power outages. She also wants to report damage to her home and easily find organizations that can help.

Frustrations: Sarah dislikes having to search through different websites and social media pages to find information. She gets frustrated when information is outdated or difficult to understand. She also worries about not knowing which organizations are trustworthy or where to go when her family needs immediate help.

Julian Parker

Age: 27

Hometown: Columbia, SC

Occupation: Construction Worker

Family: Single

Role: Volunteer



Personality Trait

Introvert --- Extrovert	More Extrovert
Thinking --- Feeling	Slightly Thinking
Sensing --- Intuition	Strongly Sensing
Judging --- Perceiving	More Perceiving

Bio: Julian is a construction worker who enjoys working with his hands, outdoor activities, and helping friends with home projects. He is creative, dependable, and likes using his skills to help others.

Goals: Julian wants to find volunteer opportunities where he can use his construction skills to help with rebuilding and cleanup. He wants to know exactly what type of work is needed, where he should go, and whether he needs to bring his own tools. He also wants to be able to choose opportunities that fit his schedule.

Frustrations: Julian dislikes signing up for volunteer work without knowing what he will be doing. He gets frustrated when organizations do not clearly explain what skills and supplies they need. He would like to travel to a location that needs volunteers not one that has plenty.

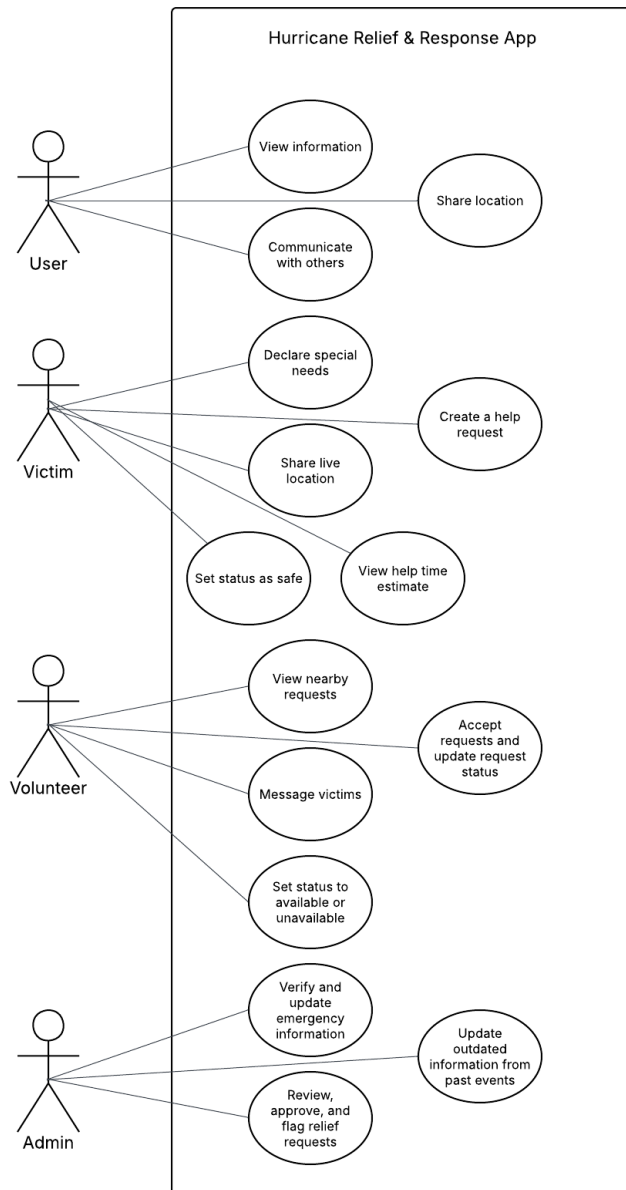
3. Constraints

1. Availability – The application should be available 24/7 for help during a hurricane or other natural disasters. It should also be working and alerting if there is one upcoming so people can prepare.
2. Response Time – Time is not on our side for hurricanes, so finding people or agents to respond quickly and to lots of people would be expensive and hard to find.
3. Connection – With bad weather, the connection could go too, so making a system that could work with limited connection would be essential, but very difficult.
4. Reliability – People depend on this for emergency information, so making sure to double check all information before posting is crucial.
5. Accessibility – The system should be easily accessible on all platform types and for different ages. It should also have a choice to speak to a person or an agent that speaks the language you are most proficient in – this could be something we use ai for, so we can get them to the correct emergency worker to talk to

4. Overall Description

This system is designed to help all types of people navigate the safest way around a hurricane or emergency caused by a hurricane. Stakeholders such as first responders, victims, volunteers, and insurance companies should all be able to use this system for their own benefit, so they can accurately help those in need. This system will provide hurricane warnings, evacuation information and locations, shelter locations, and constant updates to all. This system is meant to improve communication and help the community in such a detrimental and stressful situation and to respond to hurricanes more efficiently. The app should be user-friendly to all ages and bilingual, easy to use interface, quick responses, works without data, and accurate information.

5. Business Use Cases



6. Functional Requirements

[Requirements Spreadsheet](#)

7. Non-Functional Requirements

Look and Feel Requirements

- The user interface of the system shall be simple and easy for users to understand

Usability Requirements

- The system shall allow users to find important emergency information within a few steps
- The system shall provide clear instructions when an error occurs

Performance Requirements

- The system shall load emergency information quickly when requested

Maintainability and Support Requirements

- The system shall allow administrators to easily update information when needed

Security Requirements

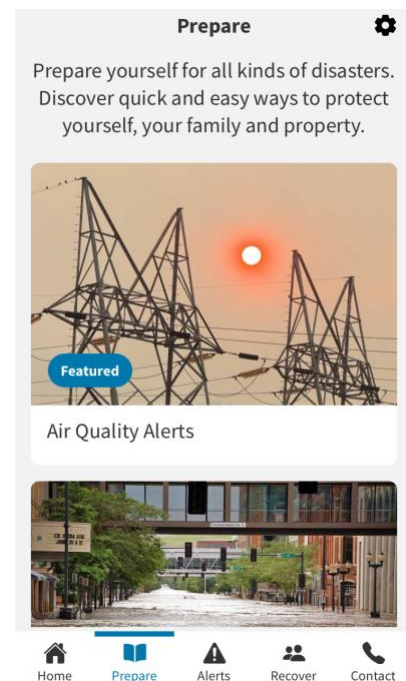
- The system shall protect users' personal information from unauthorized access
- The system shall require users to log in before accessing their personal information

Legal Requirements

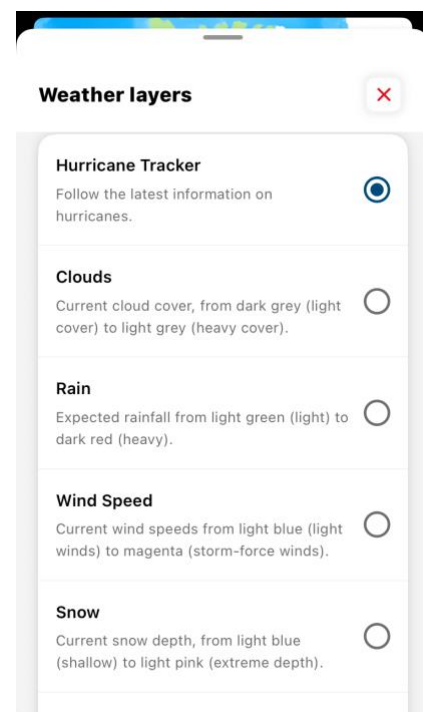
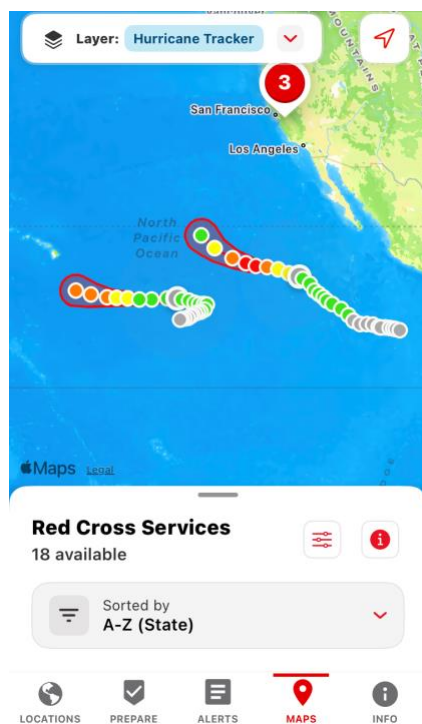
- The system shall protect users' personal information according to privacy laws
- The system shall obtain permission before sharing a user's personal information with other users.

9. Competitive Analysis

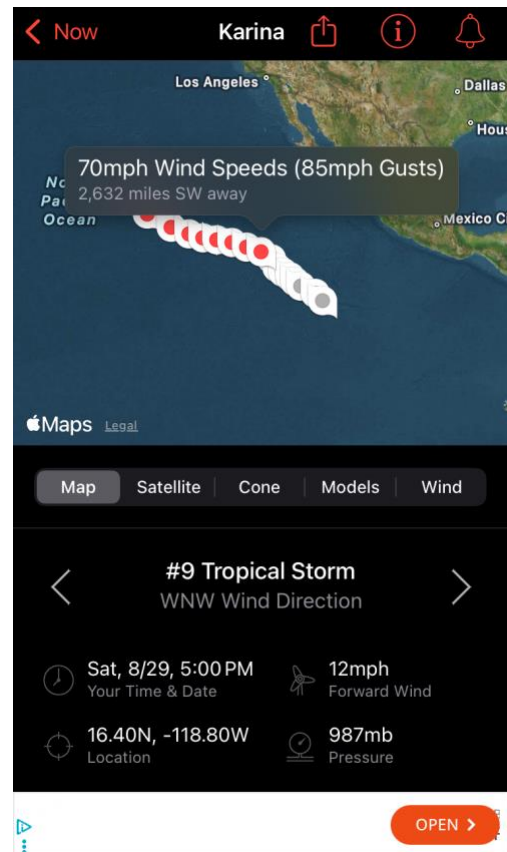
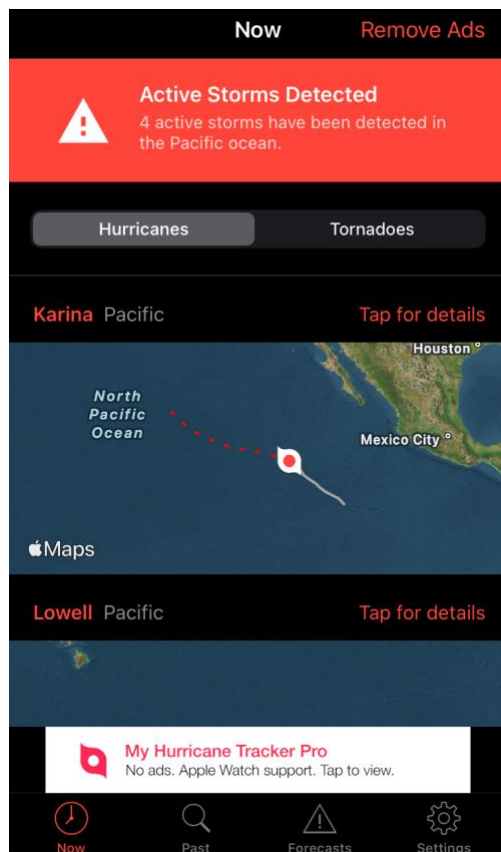
	FEMA
Strengths	• Simple UI design that makes it easy for the user to find the information they need • Has a great amount of information that is organized well ◦ A lot of information on how to prepare for disasters and recover from disasters if someone is affected by one • Users can get alerted if there is a disaster in their area ◦ Additionally, users can add multiple locations to receive alerts for • Provides a toll-free helpline for disaster survivors to call, although it has limited time availability • An official government organization
Weaknesses	• Designed more to help with any kind of disaster instead of specializing in hurricane relief • The assistance that the app provides is either before or after a disaster happens. It does not provide immediate real-time help during a disaster
Focus	The focus of FEMA is to help users prepare for any kind of disaster, notify them of disasters in their area, and provide relief for them if they are affected by a disaster. The app has a simple UI to ensure the user is easily able to find the information they need.



	American Red Cross Emergency App
Strengths	- Centralized around sending alerts for many hazards, including dust, earthquake, severe thunderstorms, and hurricanes - Each hazard has advanced alert options that allows the user to freely customize the alerts they receive - Offers live location tracking to receive real-time weather alerts - Has a lot of preparation information on each different type of hazard - Provides a map that shows Red Cross shelter locations - The map also has different weather layers to view, such as hurricane tracker, clouds, and rain - Overall extremely interactive and simple to use UI
Weaknesses	- Not much information outside of the “Prepare” tab. There is an “Info” tab, but it advertises its sponsors and the real-time weather tracking for alerts that is already shown in its home “Locations” tab - Very limited amount of Red Cross services, which makes the map feature less relevant - Does not provide immediate assistance or recovery options for those affected by disasters
Focus	The American Red Cross Emergency App focuses more on alerting its users of nearby disasters/hazards and preparing them on what to do if a disaster occurs. Although this app also focuses on each kind of disaster, it still provides good information about hurricanes, such as preparing for them and providing a live hurricane tracker.



	My Hurricane Tracker & Alerts
Strengths	- Various models of current hurricanes with dense information about the hurricane - Users can see current, past, and forecasted information about a specific hurricane - Users can easily share hurricane information to other people and turn on update notifications for a specific hurricane - Many different forecasts, such as radar, satellite, and severe warnings
Weaknesses	- Does not have any kind of help for users. Only tracks current and past hurricanes, providing mainly data about the hurricanes - UI is okay to use, but somewhat lackluster - Advertisements at the bottom near the main navigation bar, which can make it more difficult for a user to navigate around the app
Focus	The focus of this app is to track hurricanes and view data of a specific hurricane. It has many in-depth models of hurricanes, which can be useful for users who want to track hurricanes so that they can be prepared for hurricanes near them. However, this would not be the ideal app for users who need more preparation information on hurricanes, immediate assistance during a hurricane, or relief/assistance after they are affected by a hurricane.



Summary

	Strengths	Weaknesses	Focus
FEMA	+ Easy to find information for preparation and relief regarding disasters + Toll-free helpline, official government organization, and many resources	- No immediate assistance during a disaster - Not designed for hurricane specific assistance	Help people find information to prepare for disasters, recover from disasters, and get alerts for nearby disasters
American Red Cross Emergency App	+ Many customizable alert options for any disaster and hazard + Real time map with hurricane overlay tracker and live weather tracking	- No information on how to recover from a disaster - No immediate assistance during a disaster	Alerting users of nearby disasters & hazards and preparing users on what to do for each disaster & hazard
My Hurricane Tracker & Alerts	+ Tracks current hurricanes in the present, past, and future forecasts + Provides detailed models and information for each hurricane	- No preparation or recovery information for users affected by hurricanes - Lackluster UI	Track any known hurricanes and provide models and information about them

After conducting a competitive analysis on different natural disaster applications, we have found many strengths and weaknesses that will guide us to create our own hurricane relief system. We found that many of the strengths from the FEMA and American Red Cross Emergency apps are features that we aim to implement into our own app. We want users to have access to plenty of information about hurricanes, including how to prepare for a hurricane, what to do during a hurricane, and what to do if they're affected by one. Additionally, we want users to be aware of any hurricanes that are near them, so taking inspiration from American Red Cross' customizable alert options would work well in our relief system. Although these apps provide

useful information and alerts, they also share the common weakness of not having any immediate assistance during a disaster. We want to address this weakness by creating a system where users can request assistance during a hurricane and potentially connect them with available volunteers that can help them. Overall, we want to take into account the strengths and weaknesses found in this competitive analysis to create our own hurricane relief system. Our system will focus on providing users with information on how to prepare for a hurricane, what to do during a hurricane, how to request help during a hurricane, and what to do if they are affected by a hurricane. It will also send users alerts of nearby hurricanes and allow them to properly track them.